

ATIVIDADE 2 – RIAK

Implementação e Gerenciamento de Bancos de Dados NoSQL

Código no GitHub: github.com/joseniltoncirne/Implementacao-e-Geren-de-banco-de-dados/tree/main/aula05-riak/atividade02-riak

Preparando o Riak

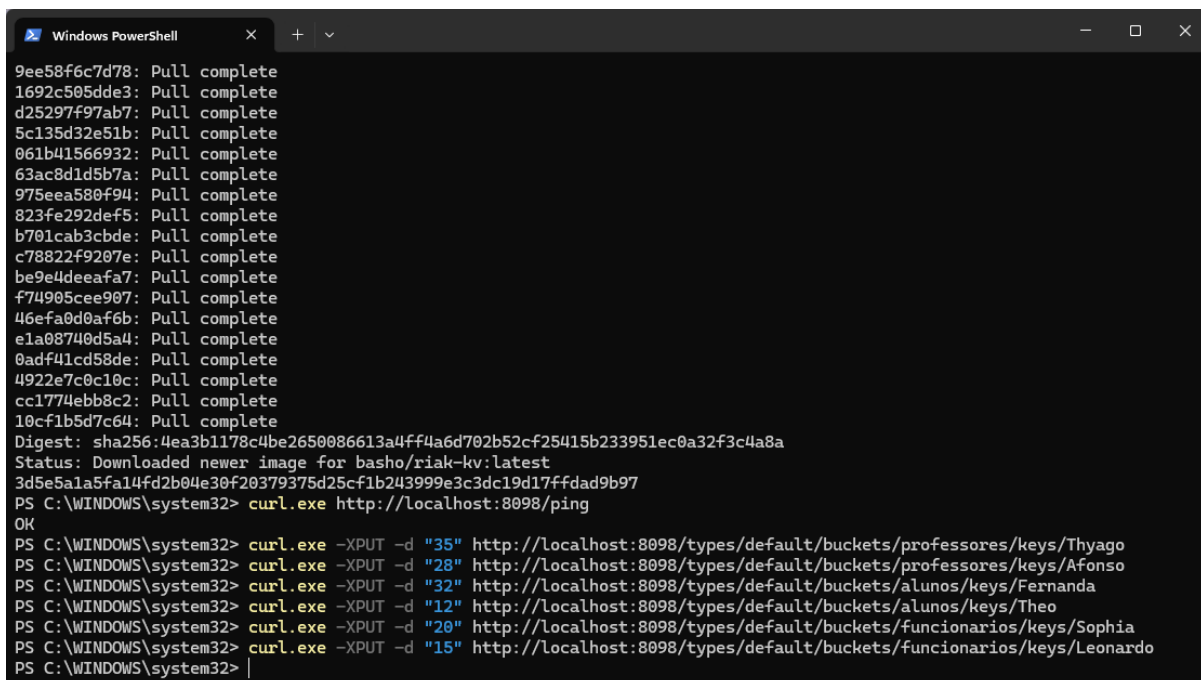
O Riak não tem instalador pra Windows, então rodei ele pelo Docker Desktop. O comando abaixo baixa a imagem e sobe o Riak com a API HTTP na porta 8098. Todos os comandos foram feitos no PowerShell com o curl.exe, e o ping respondendo OK mostra que o Riak já estava funcionando.

```
docker run -d --name riak -p 8087:8087 -p 8098:8098 basho/riak-kv
curl.exe http://localhost:8098/ping
```

Etapa 1 – Inserindo os objetos

Cada pessoa virou uma chave (o nome), com a idade como valor, dentro do bucket dela: professores, alunos ou funcionarios. Quando o PUT dá certo, o Riak não mostra nada.

```
curl.exe -XPUT -d "35" http://localhost:8098/types/default/buckets/professores/keys/Thyago
curl.exe -XPUT -d "28" http://localhost:8098/types/default/buckets/professores/keys/Afonso
curl.exe -XPUT -d "32" http://localhost:8098/types/default/buckets/alunos/keys/Fernanda
curl.exe -XPUT -d "12" http://localhost:8098/types/default/buckets/alunos/keys/Theo
curl.exe -XPUT -d "20" http://localhost:8098/types/default/buckets/funcionarios/keys/Sophia
curl.exe -XPUT -d "15" http://localhost:8098/types/default/buckets/funcionarios/keys/Leonardo
```

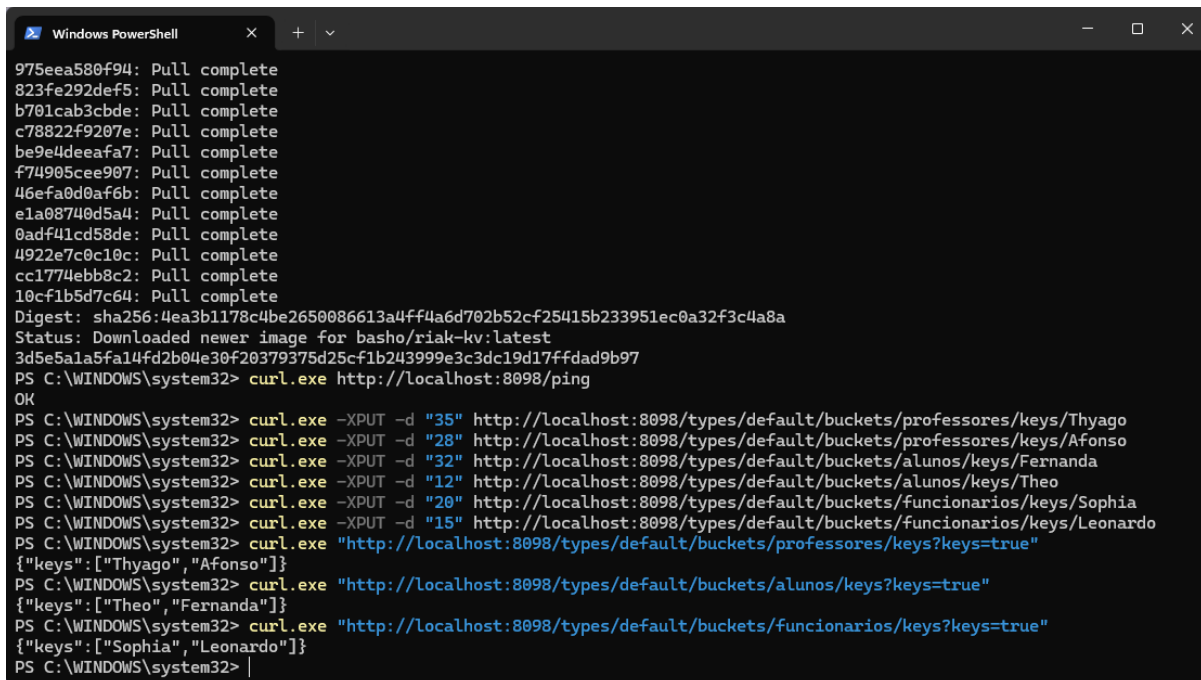


```
Windows PowerShell
9ee58f6c7d78: Pull complete
1692c505dde3: Pull complete
d25297f97ab7: Pull complete
5c135d32e51b: Pull complete
061b41566932: Pull complete
63ac8d1d5b7a: Pull complete
975eea580f94: Pull complete
823fe292def5: Pull complete
b701cab3cbde: Pull complete
c78822f9207e: Pull complete
be9e4deefa7: Pull complete
f74905cee907: Pull complete
46efa0d0af6b: Pull complete
e1a08740d5a4: Pull complete
0adf41cd58de: Pull complete
4922e7c0c10c: Pull complete
cc1774ebb8c2: Pull complete
10cf1b5d7c64: Pull complete
Digest: sha256:4ea3b1178c4be2650086613a4ff4a6d702b52cf25415b233951ec0a32f3c4a8a
Status: Downloaded newer image for basho/riak-kv:latest
3d5e5a1a5fa14fd2b04e30f20379375d25cf1b243999e3c3dc19d17ffdad9b97
PS C:\WINDOWS\system32> curl.exe http://localhost:8098/ping
OK
PS C:\WINDOWS\system32> curl.exe -XPUT -d "35" http://localhost:8098/types/default/buckets/professores/keys/Thyago
PS C:\WINDOWS\system32> curl.exe -XPUT -d "28" http://localhost:8098/types/default/buckets/professores/keys/Afonso
PS C:\WINDOWS\system32> curl.exe -XPUT -d "32" http://localhost:8098/types/default/buckets/alunos/keys/Fernanda
PS C:\WINDOWS\system32> curl.exe -XPUT -d "12" http://localhost:8098/types/default/buckets/alunos/keys/Theo
PS C:\WINDOWS\system32> curl.exe -XPUT -d "20" http://localhost:8098/types/default/buckets/funcionarios/keys/Sophia
PS C:\WINDOWS\system32> curl.exe -XPUT -d "15" http://localhost:8098/types/default/buckets/funcionarios/keys/Leonardo
PS C:\WINDOWS\system32> |
```

Etapa 1 – Listando as chaves

Aqui dá pra ver que cada bucket ficou com as duas pessoas certas.

```
curl.exe "http://localhost:8098/types/default/buckets/professores/keys?keys=true"
curl.exe "http://localhost:8098/types/default/buckets/alunos/keys?keys=true"
curl.exe "http://localhost:8098/types/default/buckets/funcionarios/keys?keys=true"
```

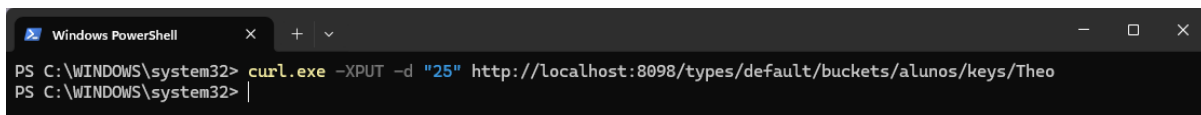


```
Windows PowerShell
975eea580f94: Pull complete
823fe292def5: Pull complete
b701cab3cbde: Pull complete
c78822f9207e: Pull complete
be9e4deefa7: Pull complete
f74905cee907: Pull complete
46efa0d0af6b: Pull complete
e1a08740d5a4: Pull complete
0adf41cd58de: Pull complete
4922e7c0c10c: Pull complete
cc1774ebb8c2: Pull complete
10cf1b5d7c64: Pull complete
Digest: sha256:4ea3b1178c4be2650086613a4ff4a6d702b52cf25415b233951ec0a32f3c4a8a
Status: Downloaded newer image for basho/riak-kv:latest
3d5e5a1a5fa14fd2b04e30f20379375d25cf1b243999e3c3dc19d17ffdad9b97
PS C:\WINDOWS\system32> curl.exe http://localhost:8098/ping
OK
PS C:\WINDOWS\system32> curl.exe -XPUT -d "35" http://localhost:8098/types/default/buckets/professores/keys/Thyago
PS C:\WINDOWS\system32> curl.exe -XPUT -d "28" http://localhost:8098/types/default/buckets/professores/keys/Afonso
PS C:\WINDOWS\system32> curl.exe -XPUT -d "32" http://localhost:8098/types/default/buckets/alunos/keys/Fernanda
PS C:\WINDOWS\system32> curl.exe -XPUT -d "12" http://localhost:8098/types/default/buckets/alunos/keys/Theo
PS C:\WINDOWS\system32> curl.exe -XPUT -d "20" http://localhost:8098/types/default/buckets/funcionarios/keys/Sophia
PS C:\WINDOWS\system32> curl.exe -XPUT -d "15" http://localhost:8098/types/default/buckets/funcionarios/keys/Leonardo
PS C:\WINDOWS\system32> curl.exe "http://localhost:8098/types/default/buckets/professores/keys?keys=true"
{"keys":["Thyago","Afonso"]}
PS C:\WINDOWS\system32> curl.exe "http://localhost:8098/types/default/buckets/alunos/keys?keys=true"
{"keys":["Theo","Fernanda"]}
PS C:\WINDOWS\system32> curl.exe "http://localhost:8098/types/default/buckets/funcionarios/keys?keys=true"
{"keys":["Sophia","Leonardo"]}
PS C:\WINDOWS\system32> |
```

Etapa 2 – Mudando a idade do Theo

No Riak, pra atualizar é só fazer o PUT de novo na mesma chave, que o valor antigo é trocado. O Theo passou de 12 pra 25.

```
curl.exe -XPUT -d "25" http://localhost:8098/types/default/buckets/alunos/keys/Theo
```

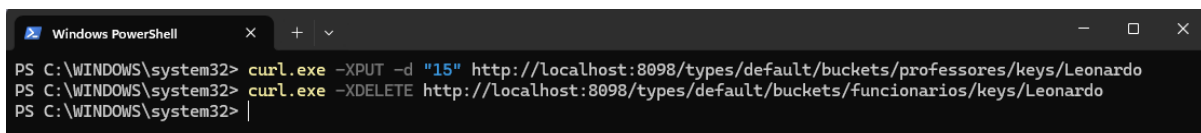


```
Windows PowerShell
PS C:\WINDOWS\system32> curl.exe -XPUT -d "25" http://localhost:8098/types/default/buckets/alunos/keys/Theo
PS C:\WINDOWS\system32> |
```

Etapa 2 – Passando o Leonardo para professores

Não tem um comando de "mover" no Riak. Então criei o Leonardo no bucket professores, com a mesma idade, e depois apaguei ele de funcionarios.

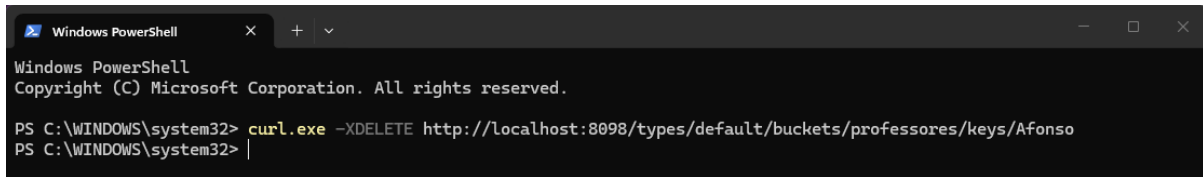
```
curl.exe -XPUT -d "15" http://localhost:8098/types/default/buckets/professores/keys/Leonardo
curl.exe -XDELETE http://localhost:8098/types/default/buckets/funcionarios/keys/Leonardo
```



```
Windows PowerShell
PS C:\WINDOWS\system32> curl.exe -XPUT -d "15" http://localhost:8098/types/default/buckets/professores/keys/Leonardo
PS C:\WINDOWS\system32> curl.exe -XDELETE http://localhost:8098/types/default/buckets/funcionarios/keys/Leonardo
PS C:\WINDOWS\system32> |
```

Etapa 2 – Apagando o Afonso

```
curl.exe -XDELETE http://localhost:8098/types/default/buckets/professores/keys/Afonso
```



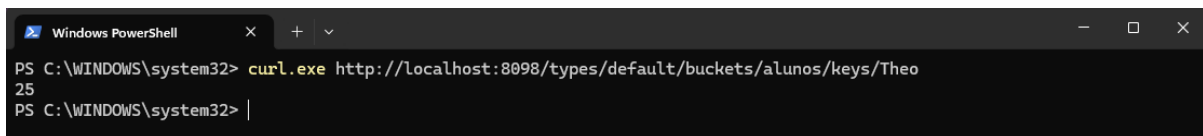
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> curl.exe -XDELETE http://localhost:8098/types/default/buckets/professores/keys/Afonso
PS C:\WINDOWS\system32> |
```

Etapa 2 – Mostrando a idade nova do Theo

O GET na chave do Theo já devolve 25.

```
curl.exe http://localhost:8098/types/default/buckets/alunos/keys/Theo
```



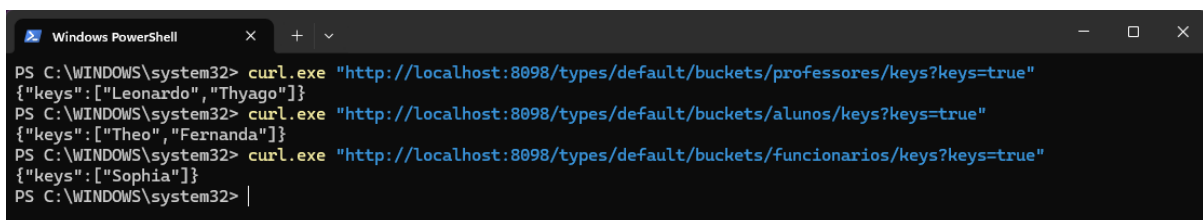
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> curl.exe http://localhost:8098/types/default/buckets/alunos/keys/Theo
25
PS C:\WINDOWS\system32> |
```

Etapa 2 – Listando as chaves de novo

No final, professores ficou com Thyago e Leonardo, alunos com Theo e Fernanda, e funcionarios só com a Sophia. O Afonso sumiu, e o Leonardo mudou de bucket.

```
curl.exe "http://localhost:8098/types/default/buckets/professores/keys?keys=true"
curl.exe "http://localhost:8098/types/default/buckets/alunos/keys?keys=true"
curl.exe "http://localhost:8098/types/default/buckets/funcionarios/keys?keys=true"
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> curl.exe "http://localhost:8098/types/default/buckets/professores/keys?keys=true"
{"keys":["Leonardo","Thyago"]}
PS C:\WINDOWS\system32> curl.exe "http://localhost:8098/types/default/buckets/alunos/keys?keys=true"
{"keys":["Theo","Fernanda"]}
PS C:\WINDOWS\system32> curl.exe "http://localhost:8098/types/default/buckets/funcionarios/keys?keys=true"
{"keys":["Sophia"]}
PS C:\WINDOWS\system32> |
```